Sports Analytics

A Dynamic Machine Learning Framework for Forecasting Test Cricket Outcomes

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The Problem: Why Test Cricket is Uniquely Hard

What makes Test different

- Spans up to 5 days / 450 overs — no ball limit
- Three outcomes: Win, Loss, **Draw**
- Pitch deteriorates — momentum shifts non-linearly
- A Draw is a strategic choice, not just a tie

Why existing models fail

- ODI/T20 models assume linear wicket value
- Logistic regression misses Draw non-linearity
- Static pre-match priors become irrelevant by day 4

899
Test matches (2001–2026)

21
Dynamic features

4
ML architectures

61.1%

Methodology: Features & Model Pipeline

Reference Team Framework

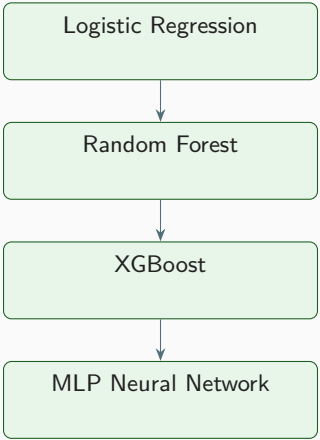
- T_{ref} = team batting first
- All 21 features centred on T_{ref}
- Target: $y \in \{-1, 0, +1\}$ (Loss/Draw/Win)

Feature Groups

Pre-match priors	Dynamic in-play
Home Factor	Match Lead (L_t)
Toss Win	Wicket Resources (WR)
Win% Diff	Relative RRR
Ground Effect	Overs Remaining
	Defensive Stance
	Follow-On flag

Non-linear wicket weights (Akhtar & Scarf 2012)

Four Architectures + Optuna Tuning



Leakage Prevention

Train/test split at **match level** — all interval

Exploratory Data Analysis

Relative Strength

- Winning teams historically stronger pre-match
- Draws cluster near zero Win% diff — evenly matched

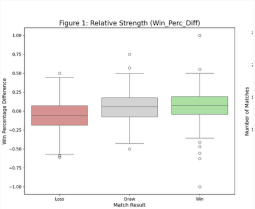


Fig 1 — Win% Diff by Result

Home Factor

- Wins more frequent **at home**
- Losses much more frequent away
- Clear home advantage in Test cricket

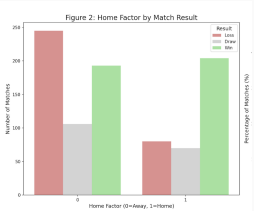


Fig 2 — Home Factor by Result

Toss Win

- Loss rate identical (40.2%) whether toss won or lost
- Toss is **statistically insignificant** in Test cricket

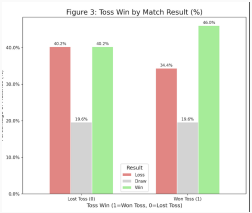


Fig 3 — Toss Win by Result (%)

EDA — The Three Decisive In-Play Signals

Match Lead (Innings 4)

- Win median ≈**290 runs**
- Draw median ≈240 runs
- Loss median ≈100 runs

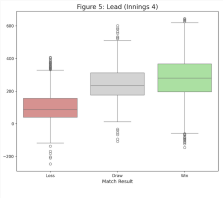


Fig 5 — Lead by Match Result

Wicket Resources

- Winners retain **more wickets** (median ≈50/100)
- Losers median ≈25
- Drives survival capacity & draw detection

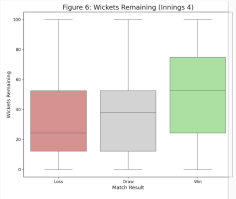


Fig 6 — Wickets Remaining by Match Result

Overs Remaining

- Losses end early (many overs unused — innings defeat)
- Draws **consume** almost all 450 overs
- Counter-intuitive: losers ≠ low overs

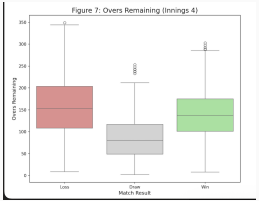
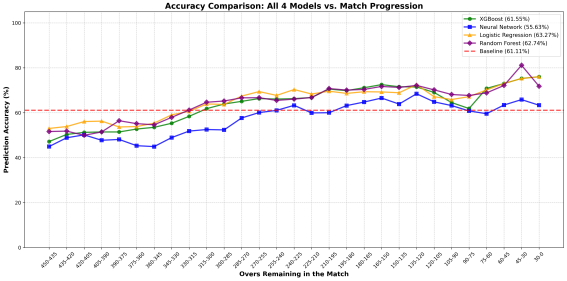


Fig 7 — Overs Remaining (Innings 4)

Model Accuracy Across Match Progression

Key observations

- Early phase (450–345 ov): all models hover 44–56%, below the 61.1% baseline — little information yet
- **Inflection** at ∼345 ov: RF, XGB, LR cross baseline and stay above it
- **Late phase** (150–0 ov): Random Forest peaks at 81% in the 45–30 over bin
- Neural Network lags 8–10 pp throughout — dataset too small for deep learning to shine on tabular data
- XGBoost and Logistic Regression finishes strongest in the *very* last bin

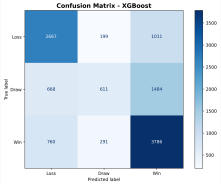


Accuracy vs. match progression — all four models vs. 61.1% baseline (dashed red)

Model Performance: XGBoost vs. Random Forest

XGBoost

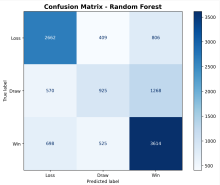
- Loss: 68.7% · Win: 78.6% · Draw: 22.2%
- 1,484 draws misclassified as Win — strong win-bias
- Lead is the #1 feature (≈0.167 gain) — 40% above #2



Confusion matrix — XGBoost

Random Forest

- Loss: 68.6% · Win: 75.1% · **Draw: 33.6%** ← highest among tree models
- Reduces draw-as-win errors by 14.5% vs XGB
- Gini impurity: Win_Perc_Diff (≈0.24) > Lead (≈0.20) — pre-match strength accumulates across all rows



Confusion matrix — Random Forest

Cross-Model Feature Importance Synthesis

Feature	XGB	RF	NN	LR
Lead	**1**	2	3	1
Win_Perc_Diff	3	1	1	9
Overs_Remaining	9	3	2	8
Ground_Effect	10	4	5	14
Ref_Wicket_Res.	5	7	6	3
Home_Factor	2	9	7	10
Toss_Win	11	11	11	13
Defensive_Stance	14	15	15	11
Runs_In_Interval	15	6	14	15

Rank order only (lower = more important); scores not comparable across models.

XGBoost feature importance (Gain metric)

Three universal findings

1 — Lead is king

Top-3 in **every** model; strongest cross-method validation of Akhtar & Scarf (2012).

2 — Toss doesn't matter

Toss_Win ranks last or near-last in all models. Coin toss provides zero advantage in Test cricket.

3 — Draw resists all models

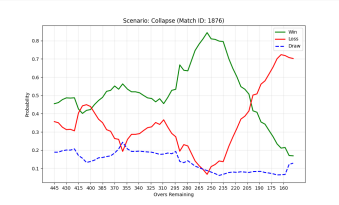
Highest misclassification across every architecture — strategic intent is inherently hard to encode numerically.

In-Play Probability Timelines (XGBoost)

Scenario 1: Collapse

Match ID 1876

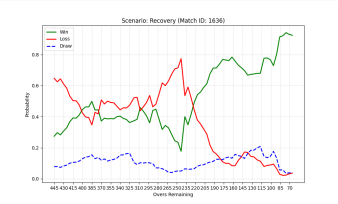
- P(Win) peaks ≈0.85 mid-match
- Rapid wicket collapse → P(Loss) climbs to 0.71 by final overs
- Model inverts prediction mid-match



Scenario 2: Recovery

Match ID 1636

- Starts at P(Loss)=0.65
- Crossover near 250 ov remaining
- P(Win) climbs to **0.93** by match end

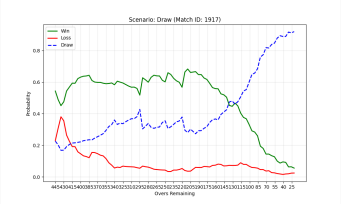


Recovery scenario

Scenario 3: Draw

Match ID 1917

- P(Win) holds 0.60–0.65 for most of match
- P(Draw) rises sharply at 130 ov remaining
- Reaches 0.93 in final 25 ov



Draw scenario

Results Summary

Overall Test-Set Accuracy

Model	Overall	Win	Loss	Draw
Logistic Reg.	**63.3%**	**78.8%**	**71.7%**	24.6%
Random Forest	62.7%	75.1%	68.6%	**33.6%**
XGBoost	61.5%	78.6%	68.7%	22.2%
Neural Net	55.6%	62.7%	59.3%	**38.6%**
Baseline	61.1%	—	—	—

Peak Late-Phase Accuracy (45–30 ov remaining)

Model	Peak Accuracy
Random Forest	**81%**
Logistic Regression	76%
XGBoost	75%
Neural Network	65%

Accuracy vs. match progression — Random Forest

Key Takeaway

Tree-based ensembles **generalise better** than deep learning on this moderate-scale tabular dataset.

Conclusions & Future Work

What we established

- ✓ **Phase transition validated**: pre-match priors fade by 300 ov; Lead absorbs all signal in final innings
- ✓ **Non-linear wicket resources** (Akhtar & Scarf) are critical for tracking momentum
- ✓ **Draw remains the hardest** outcome — product of strategic intent, time pressure, and pitch condition
- ✓ **Toss is statistically irrelevant** in Test cricket across all paradigms
- ✓ Dynamic in-play probability timelines **correctly track** Collapse, Recovery, and Draw narratives

Future directions

1 — Innings-specific models

Train four separate models (one per innings) to eliminate cross-innings generalisation overhead and reduce variance in chaotic 3rd/4th innings.

2 — Sequential deep learning

Apply **LSTM / GRU** architectures to treat each 5-over interval as a timestep, enabling the model to "remember" partnership momentum and session context.

3 — Richer features

Player-level bowling